

Project 4: Non-linear tree allometry

Overview: In this project, we will implement a nonlinear model for tree height (H) versus trunk radius (R). This model involves a classical error-in-variables model for repeated observations of the “explanatory” covariate, R , with relatively noninformative priors for the “true” or “latent” radius, μR . We will implement a model that accounts for measurement uncertainty in both the response (H) and explanatory (R) variables. The general form of the nonlinear mean function is:

$$\mu = H_{max} \left(1 - \exp \left\{ -\frac{\phi}{H_{max}} \cdot \mu R \right\} \right) \quad (1)$$

Where μR is the “true” trunk radius, H_{max} as the maximum potential height of the trees, and ϕ describes the initial rate of change (i.e., rate at which H changes with R for young, small trees).

The data: The data are H_i = height of tree i ($i = 1, 2, \dots, 184$) and $R_{i,r}$ = trunk radius of tree i measured for $r = 1, 2$, and 3 repeated observations. You may want to plot the data (H vs R) to get a better feeling for the potential relationship between H and R .

The model: To define the likelihood, we will assume that tree height is normally distributed:

$$H_i \sim N(\mu_i, \tau) \quad (2)$$

Where the mean, μ , is given by Eqn (1), and the true radius varies by tree i , μR_i . Assume that the repeated observations of trunk radius are normally distributed with a mean given by the true or latent radius of each tree:

$$R_{i,r} \sim N(\mu R_i, \tau_R) \quad (3)$$

To complete the Bayesian model, assign standard, relatively non-informative priors to the allometry parameters (H_{max} and ϕ) and the standard deviations associated with each likelihood (e.g., σ and σ_R). The precisions are computed as $\tau = \sigma^{-2}$ and $\tau_R = \sigma_R^{-2}$ (i.e., precision = 1/variance). Use the following priors:

$$\begin{aligned} H_{max}, \phi &\sim N(0, 0.00001) \\ \sigma, \sigma_R &\sim U(0, 100) \end{aligned} \quad (4)$$

The tree-specific true radii are also unknown parameters and require a prior. Use a standard (conjugate), relatively non-informative prior for the latent radii:

$$\mu R_i \sim N(0, 0.00001) \quad (5)$$

Both normal priors in Eqns (4) and (5) are parameterized in terms of a prior precision of 0.00001 (prior variance = 100,000).

Code and run the above model via R and Jags (rjags) for 3 parallel MCMC chains. You might want to provide rough estimates (initials) for the 4 parameters indicated in Eqn (4) for each of the 3 chains or see if Jags can simulate initials. Evaluate MCMC behavior (mixing and convergence) and produce posterior estimates of the model parameters.

Potential extension: Assume these trees are from a plantation where four different genotypes were planted, denoted as $g = 1, 2, 3$, or 4. The genotype of each tree is provided in the dataset. Modify the model to accommodate “genotype effects” such that the allometry parameters (H_{max} and ϕ) vary by genotype. Specify a hierarchical prior that assumes that the genotype-specific parameters arises from a normal distribution described by an overall mean (e.g., μ_ϕ) and variance or precision (e.g., τ_ϕ) that quantifies variability among genotypes. Is there evidence for genotype-specific height vs radius allometries? E.g., do the allometry parameters differ among genotypes?